STAT GR5398 Fall 2026 - Assignment 1

FinRL-Trading: a purged ML experiment and a data-coverage audit

Prepared October 11, 2026. Student: JIAO XU / jx2687. Computational assistance was used to audit code, implement and run the experiment, and prepare this draft. The student should review all claims and follow the course policy on disclosure.

1. Main result and completion status

The executed strategy returned 109.10% over 2020-06-30 to 2025-09-30, after 10 bp of cost per dollar of gross trade notional. It underperformed SPY. The requirement to beat the S&P 500 is therefore not met. These are restricted-price-universe results: absent historical securities remain in the ranked target set, but their allocation stays in cash. They are not evidence of a survivorship-bias-free full-universe strategy.

The Fall 2026 track README explicitly requires point-in-time (PIT) S&P 500 membership, SPY and QQQ comparisons, costs, a brief PDF report, runnable code/results, and a Medium link. This differs from general project descriptions that discuss NASDAQ-100, DRL, live execution, and longer-term research. None of those general descriptions has been promoted into an Assignment 1 requirement.

The online assignment and track READMEs state October 12, 2026, without an hour or timezone. Read-only Canvas checks found this course in America/New_York; assignments, modules and announcements were empty, and the pages endpoint returned 404. Thus the precise deadline remains unconfirmed. Course timezone does not prove a due-time timezone. The old Spring 2026 date inside the starter notebook is not used.

2. Inputs and PIT preparation

Official data: finrl_trading.db, fundamental_data_full.csv, and sp500_historical_constituents.csv. Fundamentals before the PIT filter: 21,299 labeled rows and 692 tickers for 2015-01-01 to 2025-09-30. After the quarter-date PIT filter: 19,007 rows, 674 tickers, 42 quarters, 2015-06-30 to 2025-09-30. The filter removed 2,292 rows (10.76%) and entirely removed 18 tickers. This is a quarter-date preparation filter; current eligibility is checked again at each decision cutoff.

Historical names ATVI, ABMD, CTXS, KSU, SIVB and FRC remain in the prepared sample. They are absent from the supplied price table, which prevents trading them in a complete historical-universe test. Historical names are retained for ranking/training rather than silently removed from the fundamental data.

3. Missing-price diagnosis

The saved original run has 174,571 selected records, 418 tickers, and dates 2020-05-01 to 2025-11-25. There are 62 selected tickers completely absent from all 1,275,978 price rows (506 tickers). They account for 18,300 original selected records, or 10.48%. This is not caused by the notebook's later date restriction.

For 2020-05-01 to 2025-09-30, 6,330 selected records fall on dates absent from the stock-price calendar; 17,650 of 163,743 selected records on existing market dates lack quotes (10.78%). For the earlier notebook's 2020-12-01 to 2025-09-30 overlap, independent rechecking reproduced 157,891 records, 6,045 on 48 absent calendar dates, and 15,601 quote gaps on market dates. These denominators differ because the windows differ.

Both database constituent snapshots (2025-10-16 and 2025-10-17) have 503 members, all with prices; the only extra priced names are BRK, SPY and QQQ. This is consistent with a later-survivor price pool, although the original collection commands/failure logs are unavailable. ANTM became ELV on June 28, 2022 (Elevance primary source), and Microsoft completed its Activision Blizzard acquisition on October 13, 2023 (Microsoft primary source). Those events illustrate heterogeneous missingness; they do not justify blindly mapping all old names or replacing ATVI prices with MSFT. BRK.B/BRK-B is an unverified share-class mapping candidate and was not changed. The full 62-name list and counts are in missing_price_tickers.csv; other event histories are unresolved.

4. Why the original trained output cannot prove an unbiased return

The original 104-minute run was preserved. All 11 sectors have six result files. Winning sector-windows: XGBoost 88, Random Forest 87, LightGBM 63. The original split uses quarter dates and then timestamps predictions with each report date. Independent inspection of every sector's first window found validation labels ending after the first signal date. For sector 20, the signal is 2020-05-01, the last validation label ends 2020-09-01, and 134 validation labels are not yet realized. Also, 2,194 prepared report dates precede standardized quarter ends, and adj_close_q was among default numeric features. Using filing dates alone does not solve those availability problems.

The original baseline uses daily adjusted open-to-close returns while a benchmark uses continuous close-to-close holdings. This omits overnight portfolio returns. Missing returns filled with zero also mask absent prices. The diagnostic file retains quoted contribution and explicitly records unpriced target weight; its partial NAV is an invalid-baseline diagnostic, not a measured full portfolio. No reference README performance is presented as a personal result. A valid reproduction of the original full baseline remains unavailable without trustworthy historical prices and corrected as-of model fitting.

5. Replacement experiment with fixed parameters and as-of rules

The new experiment pools industries and uses 23 listed accounting features. It excludes adj_close_q, PE/PB/PS and other price-derived ratios. Feature availability is conservatively max(quarter end, filing date, accepted date) + one calendar day; rows with no filing or accepted date are excluded. This deliberately delays signals with unusual fiscal/report-date conventions. It does not prove vendor fundamentals are free of restatements or that filing metadata is correct.

Labels are the published y_return values. Independently, log(next trade_price / current trade_price) matches 21,318 values within 5.0e-7. The endpoint is the next actual_tradedate in the full raw ticker history, derived before filtering; this prevents shifting over a missing PIT quarter. Twelve zero labels remain as published. Training and validation require both feature availability and label endpoint on or before the decision cutoff. Inference uses all eligible rows, including rows without realized labels, so future label presence is not an eligibility filter.

At each of 21 quarter-end cutoffs from 2020-06-30 through 2025-06-30, the last four available labeled quarters form validation; up to the preceding 16 quarters form training. Two fixed models are compared by validation MSE: Random Forest and Extra Trees, each 80 trees, depth 6, min leaf 15, max_features 0.7, seed 5398. The winner is refitted on only those already-known 20 quarters. The replacement winner counts and MSEs are saved separately from the original three-model counts. Median imputation is fitted within the training data for selection and within known fitting data for refitting. There is no optimization against backtest returns.

The most recently available fundamental row per ticker is scored among PIT members at the cutoff. The top ceil(25% of eligible tickers) is selected, with ticker order breaking prediction ties. Top 25% is a fixed assignment-baseline threshold, not a claim that it is optimal. Equal target allocation is 1 / selected count. A missing/nonpositive entry quote reserves that target allocation as zero-interest cash without reranking or renormalization. Executions occur at the next stock-calendar trading close. Shares are held through overnight periods and drift between quarterly rebalances; exiting the index midquarter does not trigger an extra rebalance. No leverage, shorting or real orders are used.

The replacement ranks 2,622 quarterly target records. 282 cannot be entered at their execution date (10.76%); mean unexecuted target allocation is 10.76%. Ranking coverage is broader than execution coverage. This defines a reproducible restricted experiment but fails the requirement for proof of a fully survivorship-free tradable universe.

6. Features and model selection

Impurity importance is averaged across the 21 refitted winning models. Leading features: asset_turnover (0.095), revenue_per_share (0.081), BPS (0.079), debt_to_capital (0.054), cash_ratio (0.051). These are within-model relevance measures, not causal effects or an independent out-of-sample attribution. Correlated features can share importance. Original fitted models were not persisted, so this is replacement-model importance, not reconstructed importance for the cached RF/XGB/GBM models.

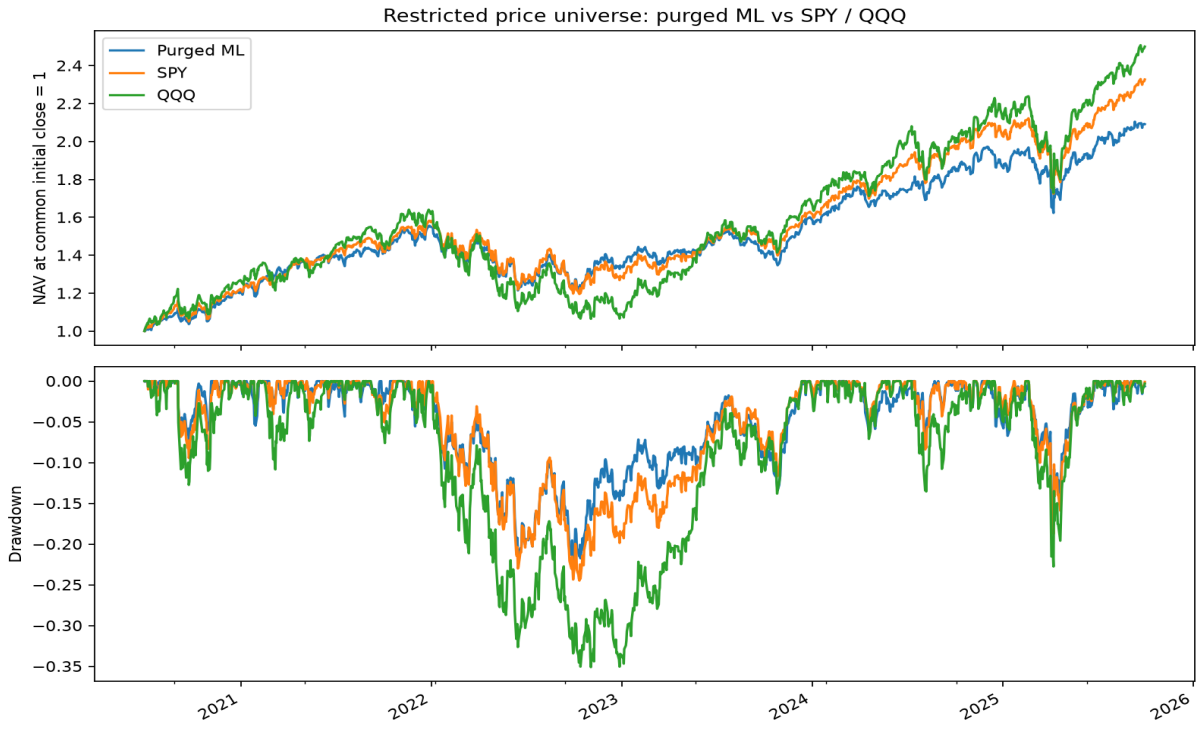
7. Portfolio results and benchmark alignment

Series	Final NAV	Total return	CAGR	Sharpe	Volatility	Max DD
Purged ML	2.091	109.10%	15.08%	0.96	15.98%	-22.22%
SPY	2.327	132.66%	17.45%	1.02	17.24%	-24.50%
QQQ	2.500	150.00%	19.06%	0.88	23.03%	-35.12%

Common initial close 2020-06-30 through 2025-09-30, 1319 daily return observations. Initial NAV is 1 before execution, so the first purchase cost is included. Stock valuations use database adjusted closes. SPY (S&P 500 proxy) and QQQ (NASDAQ-100 proxy) use explicitly downloaded Yahoo adjusted closes; the database benchmarks begin only 2020-12-01. All curves use identical dates and initial normalization. ETF adjusted returns approximate reinvested distributions, subject to provider definitions. Benchmarks are frictionless buy-and-hold; portfolio costs are charged, making the comparison conservative in that respect. Benchmark trading fees and risk-free cash interest are not included.

Portfolio mean daily gross one-way-plus-sale turnover is 0.01030; annualized mean daily turnover is 2.595. Sum of daily cost fractions relative to previous NAV is 1.36%; this is not identical to a compounded terminal wealth drag. Turnover is sum(abs(target dollar holdings - drifted dollar holdings)) / pretrade NAV. Fee is 0.001 times this turnover times NAV, including the first purchase. Fee funding reduces both stock and cash targets pro rata. No final liquidation charge is added; ending NAV marks held shares to market.

Daily return = NAV_t / NAV_(t-1) - 1; cumulative return = final NAV - 1; CAGR uses elapsed calendar days / 365.25; volatility uses sample daily SD times sqrt(252); Sharpe uses mean / sample SD times sqrt(252), with risk-free rate assumed zero; max drawdown = minimum(NAV / running peak - 1). There are no padded benchmark dates or stock quote forward fills. A missing quote for any held asset raises an error rather than creates a fabricated return.



The strategy has lower realized volatility than SPY, partly reflecting its cash allocations. Lower risk does not establish alpha, statistical significance, or superiority. Performance reflects the restricted source price pool, choice of features, fixed modeling rules, costs and dates. The reported full strategy window starts at 2020-06-30; it does not claim to reproduce the README reference period starting 2020-05-01.

8. Checks, limitations and what did not work

Executed checks: unique date-ticker price and original-selection keys; prepared PIT counts; every training and validation label endpoint <= cutoff; selected feature availability <= cutoff; target sums <= 1; finite daily returns and NAV; absence of missing held quotes; NAV equals cumprod(1 + net return). Independent final-file checks recompute all performance metrics and the cost relation. Input hashes and environment versions are recorded. Raw data and original source files were not overwritten. The 104-minute sector training was not rerun.

The original cached signals failed the as-of audit, and the new restricted strategy failed to outperform SPY. More fundamentally, missing historical security prices and unresolved corporate actions prevent the full-universe unbiased claim. The conservative signal delay and accounting-only feature subset repair specific implementation leakage channels; they cannot certify vendor PIT fundamentals, accepted-time accuracy, original label provenance, security identity, or all dividend/split/merger adjustments. The downloaded historical constituent list itself has not been reconciled to an independent index archive. These are material limitations, not minor rounding issues.

A credible next extension is to obtain authorized WRDS historical daily securities with stable identifiers and delisting/corporate-action returns, reconcile old ticker identities with effective dates, and retain terminal proceeds explicitly. Then rerun the same frozen rules before adding model/strategy variants. More variants would require time-separated validation and a sealed final evaluation period; tuning repeatedly to exceed this known SPY result would overfit. No unavailable price was fabricated, set to zero, or substituted with an acquirer price.

9. Submission and reproducibility

run_analysis.py reproduces the replacement model fit, portfolio and statistics. Cached original predictions and stock_selected.csv are included as small result files. original_code/ preserves the course preparation/training code. README.md contains data-download, preparation and execution commands. The companion notebook reads and validates saved outputs top-to-bottom; model refitting is performed by the explicit script command. Medium_draft.md is prepared but unpublished. The student must review the PDF, publish or adapt the blog, add its link, and open the course pull request themselves. No GitHub push, PR, Medium publication, email or course submission was performed.

Sources consulted (accessed October 11, 2026)

Assignment 1 requirements

FinRL track requirements

Course repository

FinRL-Trading project

General ML stock-selection framework

Elevance rebranding

Microsoft acquisition completion